

Dialectica for Constructible Falsity*

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Abstract

We can instantiate the Dialectica construction to the category of Heyting algebras, as the category of Heyting algebras has all finite products and pullbacks. Then we get a category of relations between Heyting algebras, and, in the degenerate case where the relations take values in the terminal object, a model of a logic related to Nelson’s ‘Constructible Falsity’. The Chu/Dialectica construction considered over Heyting algebras provides us with a bridge to ‘strong negation’ based logics. We discuss how, while the original idea was basically described in Patterson’s 1998 doctoral thesis, much work on Nelson’s systems was and still is required to make good on the claim that the categories can be considered models for these different logics. We also discuss briefly why this is important.¹

Dedication

This short note is dedicated to Andrea Loparic, a Brazilian logician recently departed and one of the heroines of Logic in Brazil. The original plan was doing some automated logic work with Andrea, but these plans were curtailed by her passing. Andrea (with her son Marko Loparic) had a system for proving intuitionistic theorems online. For a long time, I wanted to use their system, and especially I wanted to enrol her expertise, acquired by teaching logic from Kleene’s book [Kle52] for decades, to use automated theorem proving in a different way. More than simply proving the well-known intuitionistic theorems in Kleene’s book, I wanted to see how many different ways could we prove them true, given some restrictive assumptions. For example, what if we assumed a linear or a relevant use of hypotheses? Which one of Kleene’s basic intuitionistic theorems remains true? If they remain true, how many different proofs in sequent calculus can we find of a given theorem and how do we characterize the differences between proofs? I have not had the chance to work with Andrea on this problem. Together with Elaine Pimentel, Giselle Reis, and Carlos Olarte, we started investigating some of these questions [OPPR18]. But time is of essence in a commemorative volume, so we have decided to write down another strand of research, an application of Heyting algebras in the Dialectica construction, for this volume honouring Andrea.

*This preprint makes the paper, which appeared in a volume in memory of Andrea Loparic, findable and citable. It also corrects it: while preparing the companion work on small Dialectica categories we noticed that Section 3 conflated two levels, a Heyting algebra being a cartesian closed poset while the category **Hey** is not cartesian closed. Section 3 and Section 4 have been reworked accordingly, and the theorem about the symmetric monoidal structure has been removed. The material from Section 5 onwards works at the terminal lineale only.

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This corresponds to a simplification of the setting of the Dialectica categories to the universe of Heyting algebras, the traditional models for propositional intuitionistic logic. I hope Andrea would approve of this application. The setting of Heyting algebras makes the Dialectica construction less abstract, so we can see more clearly where certain assumptions are needed, and where they are not available. The surprise is that this setting can also be seen as a model for a variant of Nelson’s Constructible Falsity system, which always seemed to me an interesting, but flawed logic system. While the intuition behind Constructible Falsity seems very elegant, the logic does not satisfy the substitution of equals by equals, as discussed, for example, by Odintsov in [Odi04]. This seems a big problem, even if there are many interesting algebraic properties proved for the system.

1 Introduction

In this note, we describe a simplified version of the Dialectica construction [dP89b], done not over the category **Set** of sets nor over another more general category **C**, but over the specific category **Hey** of Heyting algebras.

The Dialectica construction [dP91b] is a categorical logic construction, originally based on Gödel’s Dialectica Interpretation [GF⁺86]. The construction comes in two different ‘flavors’:

- (i) an intuitionistic linear logic one, that nowadays is called Dial (as this is closer to the functionals described in Gödel’s Dialectica interpretation), published in [dP89a]; and
- (ii) a classical (or full intuitionistic) linear logic one, that used to be called GSet, *G* for Jean-Yves Girard, who suggested it as a way to provide models of classical linear logic [dP89b]. This is now called LDial, for Linear Dialectica.

Apart from the conference versions cited, both variants of the Dialectica construction are only described in the technical report of the doctoral thesis [dP91b].

The construction can be made using any category **C** with finite limits (products and pullbacks are used), but more structure is required to model the connectives of the logic. We usually describe the construction using sets, as the notion of a map in this category can be hard to grasp. The next section describes the linear version of the Dialectica construction we will be using. Then we describe the category of Heyting algebras **Hey**, which has the finite limits the construction needs. This is enough for the objects and morphisms of the Dialectica construction, and enough for everything we go on to do with Nelson’s systems, which happens at the lineale $L = 1$.

2 Dialectica constructions

The main idea of the original Dialectica construction is to build a category of relations between objects of a basic category **C**, where **C** has all finite limits. So if U and X are objects of **C**, a typical object of $\text{Dial}_2(\mathbf{C})$ will be $A \xrightarrow{\alpha} U \times X$, a subobject of the product object $U \times X$; or rather an equivalence class of monics into $U \times X$. Recall that ‘relations’ in category theory can be represented either as subobjects $A \xrightarrow{\alpha} U \times X$ or as maps into some sort of classifier like 2 . Correspondingly, we write an object A as either (U, X, α) , or $A \xrightarrow{\alpha} U \times X$, or still as $U \times X \rightarrow 2$, assuming that the category **C** has an object $2 = 1 + 1$, corresponding to truth-values $\{true = 1, false = 0\}$.

These objects intuitively represent formulae of the form $\exists u \in U \forall x \in X. \alpha(u, x)$. Thus they are not symmetric and we can think of them as games between a prover $\exists$ loise and a refuter $\forall$ belard. But more often than not we think of them as instances of problems and

possible solutions to these problems, where the relation α means that the instance of the problem is solved (or not) by a given instance of the solution.

A map between two such objects $A \xrightarrow{\alpha} U \times X$ and $B \xrightarrow{\beta} V \times Y$ is a pair of maps from $\mathbf{C}$, (f, F) , with $f: U \rightarrow V$ and $F: U \times Y \rightarrow X$, such that a non-trivial condition is satisfied. This condition is described logically in the definition below. The nature of the morphisms is that they correspond to some sort of implication between these games, or between these pairs problem/solution.

Definition 2.1 (Original Dialectica category). Given a category $\mathbf{C}$ with finite limits, the category $\text{Dial}_2(\mathbf{C})$ has as objects triples of the form (U, X, α) , sometimes written $(U \xleftarrow{\alpha} X)$, where U and X are objects of $\mathbf{C}$ and $\alpha \subseteq U \times X$ is a ‘relation’, that is, a monomorphism into $U \times X$. A map between two objects $(U \xleftarrow{\alpha} X)$ and $(V \xleftarrow{\beta} Y)$ is a pair of maps from $\mathbf{C}$, (f, F) , with $f: U \rightarrow V$ and $F: U \times Y \rightarrow X$, such that a non-trivial logical condition is satisfied, namely

$$\text{for all } u \in U \text{ and } y \in Y, \quad u \alpha F(u, y) \text{ implies } f(u) \beta y.$$

The categorical version of this condition says the following. Pull back the subobject $A \xrightarrow{\alpha} U \times X$ along the map $\langle \pi_1, F \rangle: U \times Y \rightarrow U \times X$, obtaining a subobject $A' \hookrightarrow U \times Y$:

$$\begin{array}{ccc} A' & \xrightarrow{\quad} & A \\ \alpha' \downarrow & \lrcorner & \downarrow \alpha \\ U \times Y & \xrightarrow{\langle \pi_1, F \rangle} & U \times X \end{array}$$

and pull back the subobject $B \xrightarrow{\beta} V \times Y$ along $f \times \text{id}_Y: U \times Y \rightarrow V \times Y$, obtaining a subobject $B' \hookrightarrow U \times Y$:

$$\begin{array}{ccc} B' & \xrightarrow{\quad} & B \\ \beta' \downarrow & \lrcorner & \downarrow \beta \\ U \times Y & \xrightarrow{f \times \text{id}_Y} & V \times Y \end{array}$$

The condition is then that the first subobject is smaller than the second, $A' \leq B'$ as subobjects of $U \times Y$, that is, there is a (necessarily unique) map $A' \rightarrow B'$ over $U \times Y$.

But in this short note we will be interested instead in the Linear Dialectica category, called, when it first appeared, a Dialectica-like category [dP89b]. This was suggested by Girard to simplify the original Dialectica construction, as a way of producing categorical models of Linear Logic, not only of Intuitionistic Linear Logic (ILL). Also, in what follows, we will prefer to think of the objects of this category as maps into an algebraic structure called a *lineale*. Lineales are poset reflections of symmetric monoidal closed categories, hence they are algebras with multiplicative operations tensor, negation and linear implication; see [dP02]. Clearly the lineale 2, consisting of the elements $\{0, 1\}$ meant as $0 = \text{false}$ and $1 = \text{true}$, is a specific instance of the construction below.

Definition 2.2 (Linear Dialectica category). Given a category $\mathbf{C}$ with finite limits and a lineale L in $\mathbf{C}$, the linear dialectica category $\text{LDial}_L(\mathbf{C})$ has as objects triples $(U \xleftarrow{\alpha} X)$, where U and X are objects of $\mathbf{C}$ and α , a ‘multirelation’ between U and X , is a map $\alpha: U \times X \rightarrow L$. A map between two objects $(U \xleftarrow{\alpha} X)$ and $(V \xleftarrow{\beta} Y)$ is a pair of maps from $\mathbf{C}$, (f, F) , with

$f: U \rightarrow V$ and $F: Y \rightarrow X$, such that in the diagram

$$\begin{array}{ccc} U \times Y & \xrightarrow{U \times F} & U \times X \\ f \times Y \downarrow & & \downarrow \alpha \\ V \times Y & \xrightarrow{\beta} & L \end{array}$$

we have, for all $u \in U$ and $y \in Y$,

$$\alpha(u, F(y)) \leq \beta(f(u), y) \quad \text{in } L,$$

that is, the square commutes up to the order of L , with the composite through $U \times X$ below the composite through $V \times Y$.

We could go on and describe other categorical structures available in the category $\text{LDial}_L(\mathbf{C})$, used to model the additives and especially the modalities of Linear Logic [dP91a], but we prefer to describe the category $\mathbf{C}$ we are interested in, the category of Heyting algebras.

3 Heyting algebras

Heyting algebras are the algebras used to model propositional intuitionistic logic. So they correspond, for constructive propositional logic, to Boolean algebras for classical propositional logic. One can define a Heyting algebra H as a bounded lattice with all exponential objects. Bounded means that our lattice has a top $\top$ and a bottom element $\perp$. The lattice H is regarded as a category where meet (denoted $\wedge$) is the categorical product. The exponential condition means that for any objects B and C of H an exponential C^B exists and is unique.

A Heyting implication (often written $B \rightarrow C$) is just an exponential, an alternative notation for C^B , used when we are talking about a cartesian category. From the definition of exponentials we have that implication $- \rightarrow -: H^{\text{op}} \times H \rightarrow H$ is right adjoint to meet $\wedge: H \times H \rightarrow H$. This adjunction can be written as

$$(- \wedge B) \dashv (B \rightarrow -). \quad (1)$$

This adjunction holds inside a single Heyting algebra H , regarded as a poset category: A , B and C range over the elements of H , and $\wedge$ is the categorical product of that poset. Each Heyting algebra is thus a cartesian closed poset, indeed a bicartesian closed one, since H has finite joins as well. The category **Hey** of Heyting algebras and their morphisms is a different matter: being a variety it has all finite limits, products being direct products and equalizers and pullbacks computed on underlying sets, but it is not cartesian closed, as $A \times -$ does not preserve the initial object, the free Heyting algebra on no generators. Finite limits are what Definition 2.1 and Definition 2.2 require in order to produce a category, and they are what we use.

We recall the remaining definitions for completeness.

Definition 3.1. A Heyting algebra is a residuated lattice whose monoid operation is the meet $\wedge$ of the lattice.

The monoid unit must then be the top element $\top = 1$. The commutativity of this monoid implies that the two residuals coincide, as $A \rightarrow B$.

Some examples of Heyting algebras:

1. Every Boolean algebra is a Heyting algebra, with $A \rightarrow B$ given by $\neg A \vee B$.

2. Every totally ordered set with a least element 0 and a greatest element 1 is a Heyting algebra (if viewed as a lattice). In this case $A \rightarrow B$ equals 1 when $A \leq B$, and B otherwise.
3. The simplest Heyting algebra that is not already a Boolean algebra is the totally ordered set $\{0, \frac{1}{2}, 1\}$, viewed as a lattice with the operations induced as in the previous item.

Morphisms of Heyting algebras should preserve both the distributive lattice structure and the logical implications (internal-homs). More formally:

Definition 3.2. Given two Heyting algebras H_1 and H_2 and a mapping $f: H_1 \rightarrow H_2$, we say that f is a morphism of Heyting algebras if, for any elements A and B of H_1 , we have

$$\begin{aligned} f(\perp) &= \perp, & f(A \wedge B) &= f(A) \wedge f(B), \\ f(A \vee B) &= f(A) \vee f(B), & f(A \rightarrow B) &= f(A) \rightarrow f(B). \end{aligned}$$

Using any of the equivalent definitions of Heyting algebra and the notion of morphism above, we can define the category **Hey** of Heyting algebras.

Heyting algebras are always distributive. Specifically, we always have the identities

$$\begin{aligned} A \wedge (B \vee C) &= (A \wedge B) \vee (A \wedge C), \\ A \vee (B \wedge C) &= (A \vee B) \wedge (A \vee C). \end{aligned}$$

The following infinite distributive law holds in any complete Heyting algebra:

$$A \wedge \bigvee Y = \bigvee \{A \wedge y \mid y \in Y\}$$

for any element A in H and any subset Y of H . Conversely, any complete lattice satisfying the above infinite distributive law is a complete Heyting algebra, with

$$A \rightarrow B = \bigvee \{C \mid A \wedge C \leq B\}$$

as its relative pseudo-complement operation.

4 Dialectica over Heyting

Instantiating the dialectica construction above to the category of Heyting algebras **Hey** and the lineale 2, we have:

Definition 4.1. Objects of $\text{LDial}_2(\mathbf{Hey})$ are relations $\alpha: H \times J \rightarrow 2$ between two Heyting algebras H and J , hence triples (H, J, α) . Morphisms between two objects (H_1, J_1, α) and (H_2, J_2, β) are pairs of Heyting algebra morphisms $(f, f^\#)$, where $f: H_1 \rightarrow H_2$ and $f^\#: J_2 \rightarrow J_1$ make the dialectica condition hold; this means that for all h in H_1 and all j in J_2 we have $\alpha(h, f^\#(j)) \Rightarrow \beta(f(h), j)$.

Composition in the second coordinate is not trivial, but a simple calculation allows us to state:

Proposition 4.2. *Given the category **Hey** of Heyting algebras, we can perform the dialectica construction over it, obtaining the category $\text{LDial}_2(\mathbf{Hey})$.*

This much needs only finite limits in the base, which **Hey** has.

The monoidal closed structure needs more. Both the internal-hom and the crossing tensor, written for $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ as

$$A \bowtie B = (U \times V, X^V \times Y^U, \alpha \bowtie \beta), \quad (\alpha \bowtie \beta)((u, v), (g, h)) = \alpha(u, g(v)) \wedge \beta(v, h(u)),$$

use exponentials in the base, so what they require is that **C** be cartesian closed, and **Hey** is not. They are available over a fixed Heyting algebra H regarded as a poset category, which is cartesian closed by (1). That route changes the subject matter, objects becoming triples of *elements* of H rather than triples of Heyting algebras, and we pursue it elsewhere. From the next section on, the lineale is the terminal object 1 and no exponential in the base is used.

5 Predicate-free Dialectica

So far we have provided a simple instantiation of what we had many years ago. The only difference is that we are now taking the category **C** to be the category of Heyting algebras **Hey**. However, we can also think about the dialectica objects themselves, that is, the relations $U \times X \rightarrow L$, and we can simplify these too.

Instead of the object of truth-values 2, into which we have been taking our relations, we can decide to have maps into the lineale 1, that is, some sort of degenerate relations. Since 1 is the terminal object, there is only one such unary relation, even for a generic category **C**. Moreover, the dialectica condition is then always satisfied: mapping into the terminal object 1, the square we required to commute will commute all the time, as all maps from $U \times Y$ into 1 are the same map. This construction is very degenerate. Morphisms still have to be maps of the right domain and codomain, but the dialectica condition is satisfied automatically. Hence we are talking about Chu spaces [Pra99]. The Chu condition, which requires the maps $(f, f^\#)$, $f: U \rightarrow V$, $f^\#: Y \rightarrow X$, to agree over the proposed lineale L , is vacuously satisfied when $L = 1$.

Definition 5.1. Given a dialectica category $\text{LDial}_L(\mathbf{C})$, we call the instantiation of the lineale L to the lineale 1, written $\text{LDial}_1(\mathbf{C})$, the *predicate-free* dialectica construction.

Since **C** has finite limits, it has a terminal object, so the predicate-free dialectica construction is always available, and may appear somewhat boring. After all, we intend to do some logic, but then throw it all away by making true the same as false, and collapsing all the connectives in the object 1. But this is exactly what one does when discussing compact closed categories, and some of these are very useful.

We remark that the instantiation of the lineale L to 1 can be done also in the original dialectica construction. In that case, instead of the Chu construction, one obtains lenses, or containers, but these seem to be more useful if one generalizes the system to using dependent types. There are many applications of (dependent) lenses to databases and other areas of computing.

The linear dialectica category $\text{LDial}_1(\mathbf{C})$ is both a dialectica category and a category of Chu spaces. The original dialectica category $\text{Dial}_1(\mathbf{C})$ consists of monomorphic lenses, which can be thought of as a degenerate (non-dependent version of a) polynomial category.

A dialectica-like construction for dependent type theory is given by Moss and von Glehn [MvG18], where predicates are simplified to $\top$ and the resulting category is one of polynomials. Comparisons between polynomial functors and the original dialectica categories are discussed by Dorta, Jarvis and Niu [DJN23] and by Capucci et al. [CGM⁺24], and a book-length treatment of polynomial functors is [NS25]. We do not pursue these connections here.

Similarities and differences between the categorical structures of Chu categories and those of the linear dialectica categories were initially discussed in [dP06]. There it is shown how a seemingly small change in the definition of a morphism can have a large effect on the categorical structure of the whole category. In particular, we can see that the easy tensor product of the original dialectica category, which also exists in the linear dialectica construction, is not a left adjoint to the internal-hom any more. We need to consider a much more complicated, ‘crossing’ tensor product to obtain a monoidal closed structure when we move to the setting of the linear dialectica category.

More recently, Shulman [Shu20] described a generalization of both dialectica and Chu spaces, the 2-Chu–Dialectica construction over polycategories, which allows us to discuss 2-categorical dualities and has some nice connections to multivariable adjunctions. These higher-order categorical constructions are fascinating. However, in this note we discuss a different kind of connection, a logical one, to David Nelson’s logics of strong negation and constructible falsity. We first recall Nelson’s systems.

6 Nelson’s systems

David Nelson introduced a number of systems of non-classical logics in the papers [Nel49, Nel59, AN84], providing for more fine-grained analyses of notions such as ‘falsity’ and ‘negation’ than either classical or intuitionistic logic can afford. His resulting logics combine an intuitionistic approach to truth with a dual intuitionistic treatment of falsity. His systems were investigated algebraically and using proof-theoretical tools (mainly sequent calculus and natural deduction) in many publications. A comprehensive survey is found in [KW12].

Nelson’s systems strengthen the concept of negation in intuitionistic logic in that strong negation is somewhat constructive: it produces counterexamples for negated propositions. Nelson’s analysis of the meaning of ‘falsity’ is analogous to the intuitionistic analysis of ‘truth’. The property advocated by Nelson — namely, that if a formula $\neg(A \wedge B)$ is provable, then either $\neg A$ or $\neg B$ is provable — may be regarded as dual to the well-known disjunction property of intuitionistic logic. This property is what Nelson calls ‘constructible falsity’.

Considering only the propositional fragment of Nelson’s systems, the language of the logics $N3$ and $N4$ comprises a conjunction ($\wedge$), a disjunction ($\vee$), a ‘strong’ negation ($\sim$) and two implications — a ‘weak implication’ ($\rightarrow$) and a ‘strong implication’ ($\Rightarrow$), usually introduced via the following term:

$$A \Rightarrow B := (A \rightarrow B) \wedge (\sim B \rightarrow \sim A).$$

The presence of two implications is crucial in Nelson’s logics. The strong implication in $N3$ has the feature that $(p \wedge \sim p) \Rightarrow q$ is not provable, while the weak implication satisfies the more usual $(p \wedge \sim p) \rightarrow q$. Meanwhile, the weak implication ($\rightarrow$) allows us to see $N3$ and $N4$ as conservative extensions of positive intuitionistic logic, but the strong implication ($\Rightarrow$) permits us to view their algebraic counterparts as residuated structures, and therefore to regard $N3$ and $N4$ as strengthenings of well-known substructural or relevance logics, as explained in [NRMS20].

Nelson’s logics share many properties with the positive fragment of intuitionistic logic; in particular, they do not validate Peirce’s law $((p \Rightarrow q) \Rightarrow p) \Rightarrow p$. According to Kamide and Wansing [KW12], page 2, a Hilbert-style system for $N3$ is obtained from that of $N4$ by adding the axiom scheme $(p \wedge \sim p) \rightarrow q$. In this sense, $N4$ can be seen as a paraconsistent weakening of $N3$. In fact, the logic $N3$ is the explosive extension of $N4$, i.e. the least logic in the appropriate language containing the axioms of $N4$ and the law $\sim p \rightarrow (p \rightarrow q)$, from which we can derive $N3 \vdash \perp \rightarrow p$ and $N3 \vdash \sim \perp$ (where $\perp$ is defined as $\sim(p \rightarrow p)$) [Odi05].

7 Constructible duality

We are interested here in the doctoral work of Patterson [Pat98], which provides a bridge between Nelson’s constructive systems and Dialectica/Chu spaces. Patterson recalls Nelson’s work on constructible falsity and then defines her own “constructible duality”. Constructible duality is obtained from constructible falsity by adding to the system modality operators similar to the ones of Linear Logic; see page 8 of [Pat98]. The sequent rules for constructible duality CD are found on pages 64–65 of [Pat98].

Patterson discusses some algebraic models for constructible duality, including Chu spaces and predicate-free dialectica categories as above, but does not present an explicit Curry–Howard correspondence for the system. We can consider the category $\text{LDial}_1(\mathbf{Hey})$, where objects are pairs of Heyting algebras together with a multivalued relation into $K = 1$, so triples (H, J, α) with $\alpha: H \times J \rightarrow 1$. Maps are pairs of Heyting algebra morphisms (f_0, f_1) , $f_0: H \rightarrow H_1$, $f_1: J_1 \rightarrow J$, satisfying the dialectica condition. However, as suggested above, if $K = 1$ (where 1 is the terminal object) then $\text{LDial}_1(\mathbf{Hey}) = \text{Chu}_1(\mathbf{Hey})$, so these are simply pairs of Heyting algebras, with a positive-and-negative notion of morphism. The structures of $\text{LDial}_K(\mathbf{Hey})$ and $\text{Chu}_K(\mathbf{Hey})$ are different, as explained in [dP06]; however, over the terminal object these structures become less important, since in the singleton category all logical operations are the same. It is worth noting that Shulman [Shu20, Remark 5.6] records the same collapse, and shows in addition that $L = 1$ is the *only* point at which the two constructions coincide.

The algebras considered by Patterson in [Pat98] do seem to correspond to instances of the Dialectica and Chu constructions when the ‘field’ K is the terminal object of the category in question. It would be sensible to try to provide, for the logics of constructible falsity (either $N3$, $N4$, S or CD), a Curry–Howard correspondence.

This requires a natural deduction term calculus with the appropriate properties, and this necessitates a resolution of the main problem with Nelson’s systems, described by Tranchini in [Tra21] as follows.

En somme, dans la logique $N4$, l’implication mutuelle de deux formules n’est pas une congruence, et le principe de substitution (commun à la logique intuitionniste, à la logique classique et à la plupart des systèmes logiques):

$$\frac{A \iff B}{C[A/p] \iff C[B/p]}$$

est invalide, et doit être affaibli ainsi:

$$\frac{A \iff B \quad \sim A \iff \sim B}{C[A/p] \iff C[B/p]}$$

That is, informally: in $N4$ logic the mutual implication of two formulas is not a congruence, and the principle of substitution (common to intuitionistic logic, classical logic and most logical systems) is invalid, and must be weakened by adding the second premiss $\sim A \iff \sim B$.

Thus providing a Curry–Howard interpretation for Nelson’s logics is more complicated than usual. We cannot simply follow the usual procedure as described in ‘Proofs and Types’ [GTL89], for example.

Kamide and Wansing [KW12] suggest a solution to the problem, passing through an embedding of $N4$ into substructural logics. Tranchini has a different proposal for a Curry–Howard term calculus [Tra18], which would be interesting to develop side by side with the corresponding categorical semantics. His proposal involves ‘polarized formulas’, but this requires some time and hence is left for future work.

8 Modalities for translations

When we discuss a categorical model of Intuitionistic Linear Logic (without modalities) we must provide a symmetric monoidal closed category with products and coproducts. This is sometimes called a linear category.

However, if we do have modalities in the model, we should end up with a (symmetric monoidal) adjunction between the original linear category $\mathbf{L}$ and the cartesian closed category $\mathbf{C}$ which models propositional intuitionistic logic. The adjunction models the modality ‘!’, Girard’s “of course!”. The characteristics of the modality match very closely Girard’s translation [Gir87]. And we know that, while cartesian (closed or not) structures are unique up to isomorphism, monoidal closed structures are not unique. Similarly, the modalities themselves are not determined: there may be several different (not equiprovable) modalities on the same category. The traditional example is the category of relations. This has more than one usual modality: both the finite multisets and simply the full powerset can be used as a modality, as explained in https://www.pls-lab.org/en/Relational_model_of_Linear_Logic.

One might want to consider Nelson’s strong negation in this context of modalities and translations between models, as is done in [PC00].

In the category $\mathbf{LDial}_1(\mathbf{C})$ we could consider a modality such that

$$!A = A \& I = (U \times 1, X + 1, \alpha^*)$$

where $I = (1, 1, \text{id}: 1 \times 1 \rightarrow 1)$ and $\&$ denotes the categorical product, the product of $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ being $(U \times V, X + Y, \alpha^*)$; at the lineale 1 every relation is the unique map into the terminal object. What this asks of the base is a coproduct $X + Y$ that behaves as it does in **Set**.

The idea of a simplified modality !, which is just a product with the identity for tensor, has been considered several times before. In [PC00] one asks in addition for a map $A \multimap I$ and for $I^\perp = I$. The second is easy, the linear negation here being simply swapping coordinates. The first needs a map $1 \rightarrow X$, and over **Hey** there is none for nontrivial X , since the terminal Heyting algebra has $\perp = \top$; for the same reason $X + 1 \cong 1$ there. So the theorem below is available over bases where the terminal object behaves as in **Set**, and modalities for constructible duality over **Hey** itself remain to be found.

Theorem 8.1 (Patterson 2000). *The categorical model of constructible duality is also a model of classical linear logic. Let A be a formula of Linear Logic. If A is valid in (classical) linear logic, then $\overline{A}$ is valid in constructible duality, where $\overline{A}$ is the translation described in Table 1.*

Note that Theorem 8.1 lives entirely at $L = 1$: the linear logic structure invoked here is the one available in the predicate-free setting, where the Chu and dialectica constructions agree.

9 Conclusion

We have discussed briefly how we can provide categorical semantics of some sort for Nelson’s system of constructible duality, based on Patterson’s work and on the dialectica construction. Patterson used Chu spaces mostly, but in the case discussed here, where relations are evaluated in 1, there is no difference between Chu spaces and dialectica ones.

To conclude, we want to mention why we think the Curry–Howard correspondence for (some of) the Nelson systems is important. In many practical situations, both in natural language and in more abstract settings, we have a situation where counterexamples are easier to understand than the traditional intuitionistic negation with its eventual implication into a falsity. If we want logics that agree with our intuitions, we want systems that can cope

Linear Logic	Constructible Duality
0	0
$\top$	1
I	I
$\perp$	I
$A \otimes B$	$\neg((A \supset \neg B) \wedge (B \supset \neg A))$
$A \multimap B$	$(A \supset B) \wedge (\neg B \supset \neg A)$
$A \wp B$	$(\neg A \supset B) \wedge (\neg B \supset A)$
$A \& B$	$A \wedge B$
$A \oplus B$	$A \vee B$
$A^\perp$	$\neg A$
$!A$	$A \wedge I$
$?A$	$A \vee I$

Table 1: Translating Linear Logic to Constructible Duality

with the naturality of these basic negations, at the same time as having the canonical logical properties that we have come to expect as necessary from logical systems: syntax and semantics aligned to provide soundness and completeness theorems, syntactic cut elimination and/or normalization, some sort of subformula property, etc. It would be good if the extra benefits pointed out by implicit programming [Pat98] and its use of fixed points could be harnessed as well; but even without this extra computational layer, having a Nelson system that actually provided a constructive negation, even at the price of two implications (one weak, one strong), seems a goal worth working for.

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